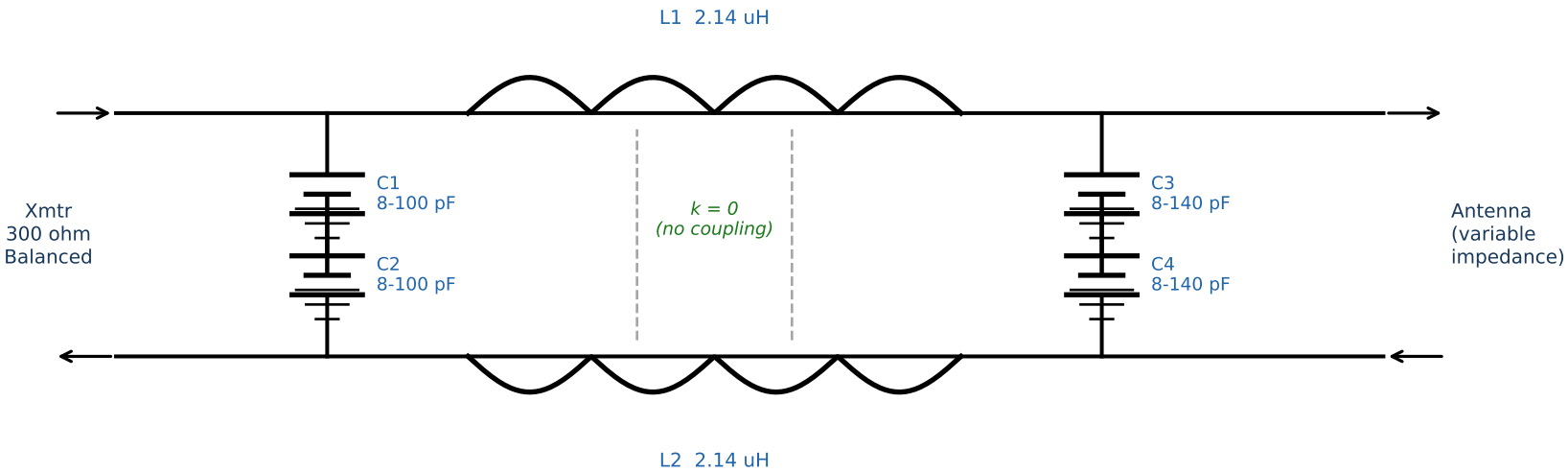


ATU Matching Range Analysis

20m CW Transmitter | Balanced Pi-Network ATU Analysis | 14.2 MHz



Coupling Constant Effect on Differential Inductance (L = 2.14 uH each, f = 14.2 MHz)

k	Effective XL (diff.)	% of k=0	Assessment
0.99 (original)	3.8 ohm	1%	Essentially zero — matching impossible
0.90	38.2 ohm	10%	Severely limited matching range
0.00 (recommended)	382 ohm	100%	Full matching range available

Why k = 0 is required:

For a balanced (push-pull) circuit the two inductors carry equal and opposite currents. The effective differential inductance is $2L(1-k)$, so a high coupling constant drastically reduces the series reactance available for impedance matching.

With $k=0.99$ the inductors act as near-perfect transformers and cancel each other differentially, leaving only 3.8 ohm of series reactance — far too little for any useful matching. The inductors must not be magnetically coupled: wind them on separate cores or orient at 90 deg.

Component Reactances and Resistive Matching Range

20m CW Transmitter — Balanced Pi-Network ATU Analysis | 14.2 MHz

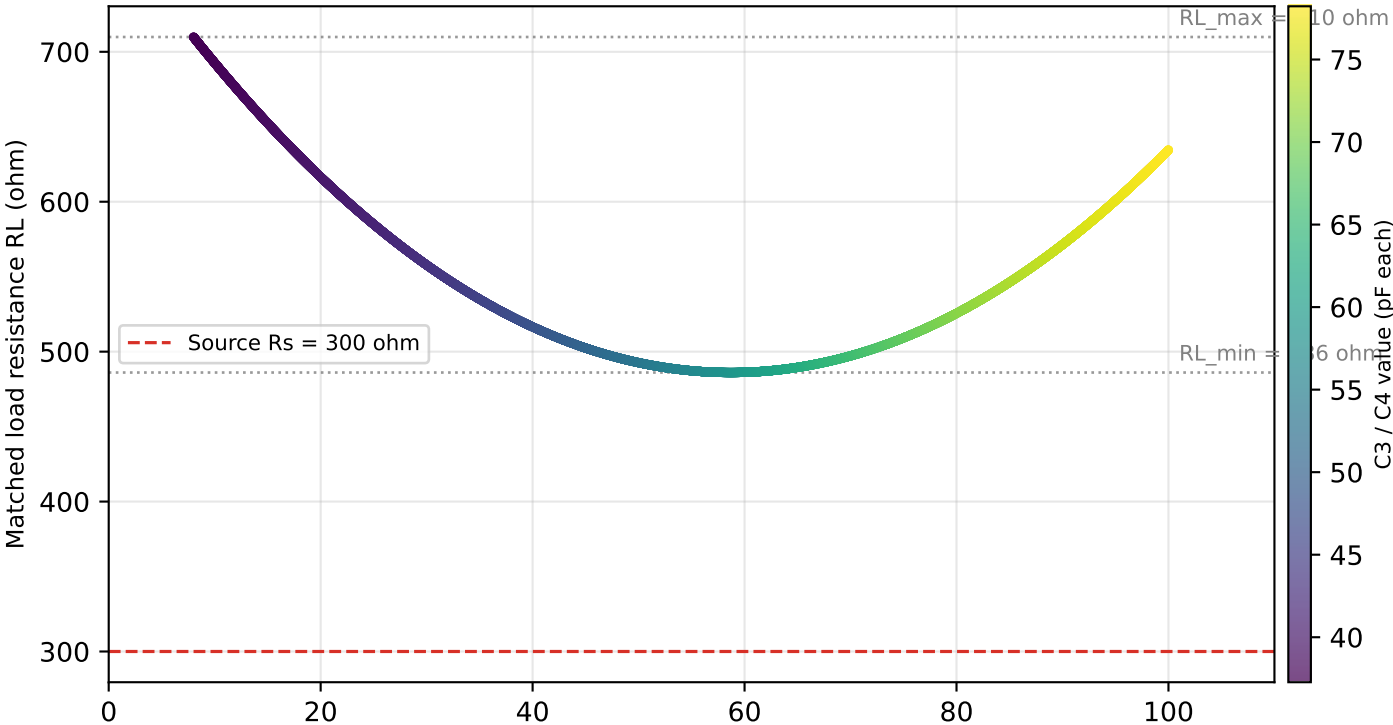
Differential Reactances at 14.2 MHz (k = 0)

Component setting	XC / XL	Role
C1/C2 @ 8 pF each	2802 ohm	Min. shunt susceptance (in
C1/C2 @ 100 pF each	224 ohm	Max. shunt susceptance (ir
C3/C4 @ 8 pF each	2802 ohm	Min. shunt susceptance (ou
C3/C4 @ 140 pF each	160 ohm	Max. shunt susceptance (o
L1 + L2 (k=0)	382 ohm	Series reactance (both indu

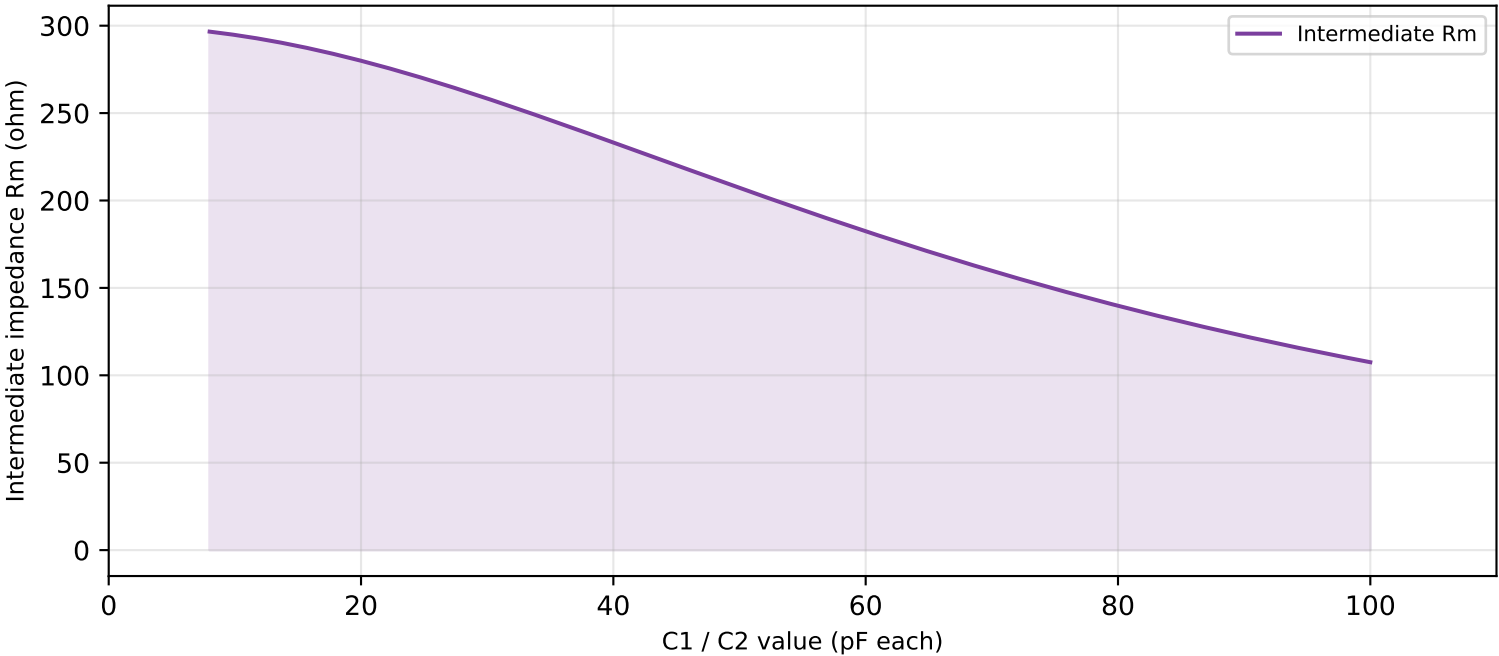
Pi-Network Design Equations

$$\begin{aligned} R_m &= R_s / (1 + Q_1^2) \\ R_L &= R_m \times (1 + Q_2^2) \\ X_L &= (Q_1 + Q_2) \times R_m \\ Q_1 &= R_s / X_{C_in} \\ Q_2 &= R_L / X_{C_out} \\ R_s &= 300 \text{ ohm, } X_L = 382 \text{ ohm} \end{aligned}$$

Resistive Load Matching Range vs. Input Capacitor Setting



Intermediate Impedance R_m vs. Input Capacitor Setting



Reactive Load Range and Design Summary

20m CW Transmitter — Balanced Pi-Network ATU Analysis | 14.2 MHz

Approximate Tunable Load Reactance Range

Inductive limit (+160 ohm): set by the maximum output capacitance (140 pF each). The output cap must resonate the inductive load component. Larger output caps extend this limit.

Capacitive limit (-382 ohm): set by the total series inductance (XL = 382 ohm, k=0). The inductor resonates a capacitive load; larger inductors extend this limit. Both bounds are conservative — the actual tunable range for complex loads is somewhat wider because the resistive and reactive parts interact.

Parameter	Value / Range
Frequency	14.2 MHz (20m CW)
Source impedance	300 ohm balanced
Each inductor	2.14 uH -> XL = 191 ohm per inductor
Series XL (k=0, diff.)	382 ohm (both inductors, push-pull drive)
Input caps C1/C2	8 pF - 100 pF each -> XC_diff: 224 - 2800 ohm
Output caps C3/C4	8 pF - 140 pF each -> XC_diff: 160 - 2800 ohm
Coupling constant	k = 0 (physically separate, non-coupled inductors)
Resistive load range	486 - 710 ohm (always steps UP from 300 ohm source)
Max inductive load X	+160 ohm (output cap limited)
Max capacitive load X	-382 ohm (series inductor limited)

Observations: The network steps the 300 ohm source UP to 486 - 710 ohm for purely resistive loads. It cannot match loads below ~486 ohm because the series inductance (382 ohm > Rs = 300 ohm) forces a minimum network Q of ~1.1. For a near-resonant folded dipole (~300 ohm +/- small X) this ATU works well. For low-impedance loads (short antenna, ground-mounted vertical) a different topology or smaller inductors would be needed. The inductive range (+160 ohm) may be limiting if the antenna operates well below resonance; larger output capacitors would extend it.